Technische Universität Braunschweig

Institut für Softwaretechnik und Fahrzeuginformatik

Official
Temporary Thesis
Title Image

My Thesis with a Reaaaaally Reaaaaally Reaaaaally Long Title

Insert Your Name

Bachelor’s Thesis

Institute of Software Engineering and Automotive Informatics

Bachelor's Thesis in Computer Science

Course of Study:	Computer Science
Student Number:	[1234567]
Date of Submission:	May 12, 2026
Institution:	TU Braunschweig

Examiners

- Prof. Dr. Thomas Thüm
  Institute of Software Engineering and Automotive Informatics
- Your Second Examiner
  Their Institute

Supervisor

- First Supervisor
  Institute of Software Engineering and Automotive Informatics

Abstract

The abstract gives a short overview of your entire thesis. It can either be written as a unstructured (150-300 words, one single paragraph) or a structured (longer, fixed headlines, example below) abstract.

Context of your thesis. CONTEXT

Research objective you plan to achieve. OBJECTIVE

Methodology you use to achieve your research objective. METHOD

Limitations of your research / threats to validity LIMITATIONS

Your most central findings RESULTS

Acknowledgements

This section is optional! In the acknowledgements section you can thank all people and organizations involved in the writing process of your thesis, e.g., your family, friends that proofread your thesis, …

Contents

Chapter 3:

Why did we choose X instead of Y?

Why do we modify algorithm C?

Chapter 4:

Why did we implement algorithm B in that way?

Chapter 5:

Is algorithm B better than algorithm C?

List of Figures

List of Tables

List of Listings

1 Introduction

1.1 Motivation

The introduction motivates the research topic and gives an overview over the related scientific work using citations [2, 1]. The introduction starts with the significance of the general research area and then leads reader attention stepwise to the area of the specific research problem of the thesis. That means that the introduction is structured like an inverted triangle, starting with the general research area and then focusing more and more on the specific context of the thesis.

[2] Thüm et al. (2009), Reasoning About Edits to Feature Models.
[1] Sundermann et al. (2024), Collecting Feature Models from the Literature: A Comprehensive Dataset for Benchmarking.

⇒ Why is your research important?

1.2 Contributions

The contributions section lists all scientific contributions of your work (not the findings). This naturally includes answering your research problem/questions, but can also include reusable data sets, newly created tools, …

⇒ What does your thesis contribute to science?

1.3 Overview

A short overview over the general structure of your thesis.

Background

2

Use this summary environment to give short summaries at the beginning of each chapter. In Section 2.1, we . . . . In Section 2.1, we . . .

Use this chapter to introduce relevant parts of your topic in a way so that the average computer science reader understands it.

2.1 Background Area A

Descriptions, definitions, examples, and figures

2.2 Background Area B

3 Concept

Describe the conceptual work you did for your thesis.

3.1 Thesis Commands Examples

```
x ← list(i='hey', j='there')
for(i in x) {
   print(paste("x$i", x$i))
}
print(x)
```

To interpret the program above, we take `x` ← 2 to the set of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{R}$ because they describe funny numbers that we can use to index what we call an AST (if you want the long form use `\glsac`: Abstract Syntax Tree (AST)). Now we can use a CPO or verify the behavioral equivalence [first use!] with external factors when interpreting the code as a Model (see behavioral equivalence, sometimes called foo bar). $\mathbb{N}$ will not be repeated on this page!

Using the `\glssym` macro defined in the preamble you can even color symbol uses such as $\mathbb{N}$ automatically if this is to your liking:

$\mathbb{N}$: Set of non-negative ints.
$\mathbb{Z}$: Set of integers.
$\mathbb{R}$: Set of real numbers.
AST: Abstract Syntax Tree
CPO: Complete Partial Order
Behavioral Equivalence: All *variables* in common evaluate to the same value for all external factors.
External Factor: Factors outside of a models control
Model: Set of model elements.

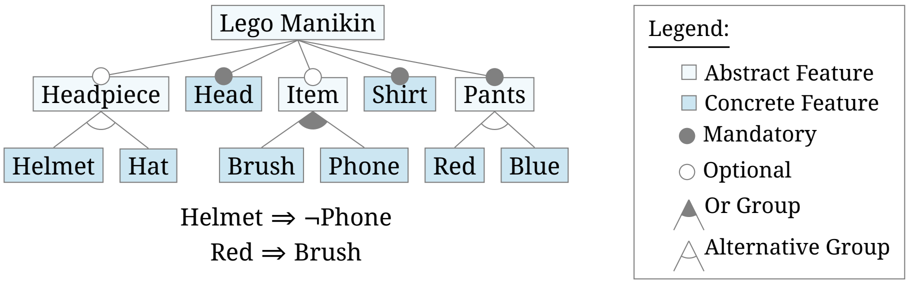
$$\mathbb{N} = \{1, 2, \ldots\} \tag{3.1}$$
$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-1, -2, \ldots\} \tag{3.2}$$
$$\mathbb{R} = \textit{Set of real numbers} \tag{3.3}$$
$$\mathbb{C} = \textit{Set of complex numbers} \tag{3.4}$$
$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R} \tag{3.5}$$

However, as you may see, $\mathbb{C}$ is not shown in the sidebar even though it appears first within the math mode. The reason(s) for this are manifold but in short we avoided this as math mode may be rendered multiple times. You can use `\showsymbols` to propose a list of symbols to show before the math environment (using `\showsymbols*` these will be forced as showcased below):

$\mathbb{C}$: Set of complex numbers.

$$\mathbb{N} = \{1, 2, \ldots\} \tag{3.6}$$
$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-1, -2, \ldots\} \tag{3.7}$$
$$\mathbb{R} = \textit{Set of real numbers} \tag{3.8}$$
$$\mathbb{C} = \textit{Set of complex numbers} \tag{3.9}$$
$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R} \tag{3.10}$$

$\mathbb{N}$: Set of non-negative ints.
$\mathbb{Z}$: Set of integers.
$\mathbb{R}$: Set of real numbers.
$\mathbb{C}$: Set of complex numbers.



(a) *Exemplary feature model. This illustrates the usage of the feature model*

(b) *Second subfigure.*

Figure 3.1: *Main caption.*

Table 3.1: *A simple table*

A	B
1	2
3	4

A viable way to group different figures is the *subfigures* command, a short example can be seen at Figure 3.1. This is Figure 3.1 with the name Main.[1] See ulm university[2] for more information.

🔗 Figure 3.1 on this page.

🔗 Figure 3.1 on this page.

1. `www.google.com` (December 24, 2024)

2. https://www.uni-ulm.de/ (December 21, 2024)

In the Algo Good algorithm Alg. 3.1, an important line is line 1.

🔗 Alg. 3.1 on this page.

You can also reference default enumerate elements:

1. First item
2. Second item

With ref: “1”, on page 4, and cref: “1”.

🔗 1 on this page.

Algorithm 3.1: *Algo Good*

In: x :: String
Out: y :: String
Out: z :: String
1 **if** x **is** "hello" **then** **return** "world"
2 **while** true **do**
3   **if** y **is** "world" **then**
4   **return** "hello"
5 **return** "goodbye"

Listing 3.1: *A floating caption*

```
x ← list(i='hey', j='there')
for(i in x) {
   print(paste("x$i", x$i))
}
print(x)
```

But please, if there is any sense in doing so make use of, e.g., `ref` by `enumitem`:

1. First item
2. Second item

With ref: “Penguin Element 1”, on page 5, and cref: “Penguin Element 1”. Using `\newlist` you can also define your own lists, e.g., for research questions or contributions and give them your own ref names — see, e.g., `rqs` for an example.

🔗 Penguin Element 1 on this page.

We can reference code as well, see: Listing 3.1.

🔗 Listing 3.1 on this page.

This starts the paragraph. “ This paragraph symbolizes text generated with a type 2 AI tool that you want to use in your thesis. Please refer to the file `segments/ai-entries.tex` for section-wide and type 1 AI disclaimers. ”[3] This continues the paragraph afterwards

3. AI quote 3.1, see page 12

4 *Implementation*

Describe the implementation work that you (maybe) performed for your thesis.

5 Evaluation

We did…

5.1 Research Questions

What are your central research questions?

RQ$_1$ What is the concept of this thesis?

RQ$_2$ How is the concept implemented?

RQ$_3$ How is the implementation evaluated?

- RQ$_{3.1}$ How is the correctness evaluated?
  - RQ$_{3.1.1}$ Please do not nest the questions that deep!
- RQ$_{3.2}$ How is the performance evaluated?

We can now reference RQ$_1$ or with cleveref: Research Question RQ$_1$.

Research Question RQ$_1$ on this page.

We can also reference sections, e.g., Section 5.1

Section 5.1 on this page.

5.2 Method

How did you perform your experiments?

5.3 Results

What are the results of your experiments?

5.4 Discussion

What are the answers to your research questions?

5.5 Threats to Validity

What threatens the validity of your findings? Discuss at least internal and external validity, possibly more [3].

[3] Wohlin et al. (2012), Experimentation in Software Engineering.

5.6 Summary

Short summary of this section

6 Related Work

In the related work section, you present other research that is relevant for your research domain. Structure other works by topic, give short descriptions, and argue why your research adds to the already existing literature (why it is different).

7 Conclusion

Within this chapter, we summarize the work of …

7.1 Summary

Short summary of your thesis, especially your findings.

7.2 Future Work

Research areas that have now opened because of your work or improvements you would want to perform.

Bibliography

[1] Chico Sundermann et al. “Collecting Feature Models from the Literature: A Comprehensive Dataset for Benchmarking”. In: *Proc. Int’l Systems and Software Product Line Conf. (SPLC)*. Dommeldange, Luxembourg: ACM, Sept. 2024, pp. 54–65. DOI: 10.1145/3646548.3672590. On page 1.

[2] Thomas Thüm, Don Batory, and Christian Kästner. “Reasoning About Edits to Feature Models”. In: *Proc. Int’l Conf. on Software Engineering (ICSE)*. Vancouver, Canada: IEEE, May 2009, pp. 254–264. ISBN: 978-1-4244-3453-4. DOI: 10.1109/ICSE.2009.5070526. On page 1.

[3] Claes Wohlin et al. *Experimentation in Software Engineering*. Berlin, Heidelberg: Springer, 2012. ISBN: 978-3-642-29043-5. DOI: 10.1007/978-3-642-29044-2. On page 7.

Glossary and Acronyms

Glossary

B

Behavioral equivalence Two models A and B have the same observable behavior for all *variables* they do have in common… On page: 3.

E

External factor A factor defined outside of the model and its expressions. Its change can yield completely different values inferred by the solver. On page: 3.

M

Model Finite set of model elements … On pages: 3, 11.

Model element Component of a model … On pages: 3, 11.

Acronyms

A

AST Abstract Syntax Tree On page: 3.

C

CPO Complete Partial Order On page: 3.

Notations

C

Complex ($\mathbb{C}$) Describes the set of complex numbers. On page: 3.

I

Integer ($\mathbb{Z}$) Describes the set of non negative as well as the negative integers: $\{\ldots, -1, 0, 1, 2, \ldots\}$. On page: 3.

N

Natural Number ($\mathbb{N}$) Describes the set of non negative integers: $\{1, 2, \ldots\}$. If 0 is to be included, it is indexed as $\mathbb{N}_0$. On page: 3.

R

Real number ($\mathbb{R}$) Describes the set of real numbers. On page: 3.

AI Usage

While writing the submitted Bachelor's Thesis, I followed the ISF's current guidelines on the use of AI. I have listed all instances of AI use in the following section, as described in the guidelines. I have also documented any use of AI outside of the writing of this thesis, e.g., for programming or information gathering, as described in the guidelines.

Jert P.

Braunschweig, December 30, 1649

Location, Date

Insert Your Name

Instance Listing

Type 1 AI Systems

Tool:	DeepL.com (free version). Accessed: 07.04.2026
Used in:	Section 1.2 and Chapter 3
History/Prompt:	Translation from German to English
Sentiment:	The tool translated the original text really well. I did not need to change something.

Section 1.2 on page 1.
Chapter 3 on page 3.

Type 2 AI Systems

Tool:	GPT-5.2
Used in:	UserController.java
History/Prompt:	Generate a Spring Boot REST controller with CRUD endpoints.
Sentiment:	It generated code that almost gave me the correct result. I tried to change my prompt so that it would fix the problem, but ultimately had to make some manual corrections by hand.
Tool:	GPT-5.2
Used in:	AI quote 3.1
History/Prompt:	Generate a text snippets that symbolizes AI usage based on the ISF AI guidelines.
Sentiment:	Worked quite well!

AI quote 3.1 on page 5

Declaration of Authenticity

I, Insert Your Name, hereby declare that this bachelor thesis is my own unaided work. All direct or indirect sources used are acknowledged and referenced. This bachelor thesis was not previously presented to another examination board and has not been published as of yet.

Braunschweig, December 30, 1649	Jert P.
Location, Date	Insert Your Name

Task Description

The following pages contain the task description (proposal) for the submitted Bachelor's Thesis. The included version was accepted by the supervising professors as task description for the presented Bachelor's Thesis.

Braunschweig, *December 30, 1649* — Jert P.

Location, Date — Insert Your Name

Proposal — Example Title

Bachelor's Thesis Proposal

Author:	Author Name
Advisor:	M.Sc. Your Advisor
Reviewer:	Prof. Dr. Thomas Thüm
Planned Registration:	Planned Registration Date

1 Introduction

In one to two pages, the introduction motivates the research topic and gives an overview over the related scientific work using citations [TBK09]. The introduction starts with the significance of the general research area and then leads reader attention step-wise to the area of the specific research problem of the thesis. That means that the introduction is structured like an inverted triangle, starting with the general research area and then focusing more and more on the specific context of the thesis.
⇒ Why is your research important?

2 Problem Statement

In three to four sentences, the problem statement describes the research problem the thesis aims to address.
⇒ What problem does your research plan to solve?

3 Contribution

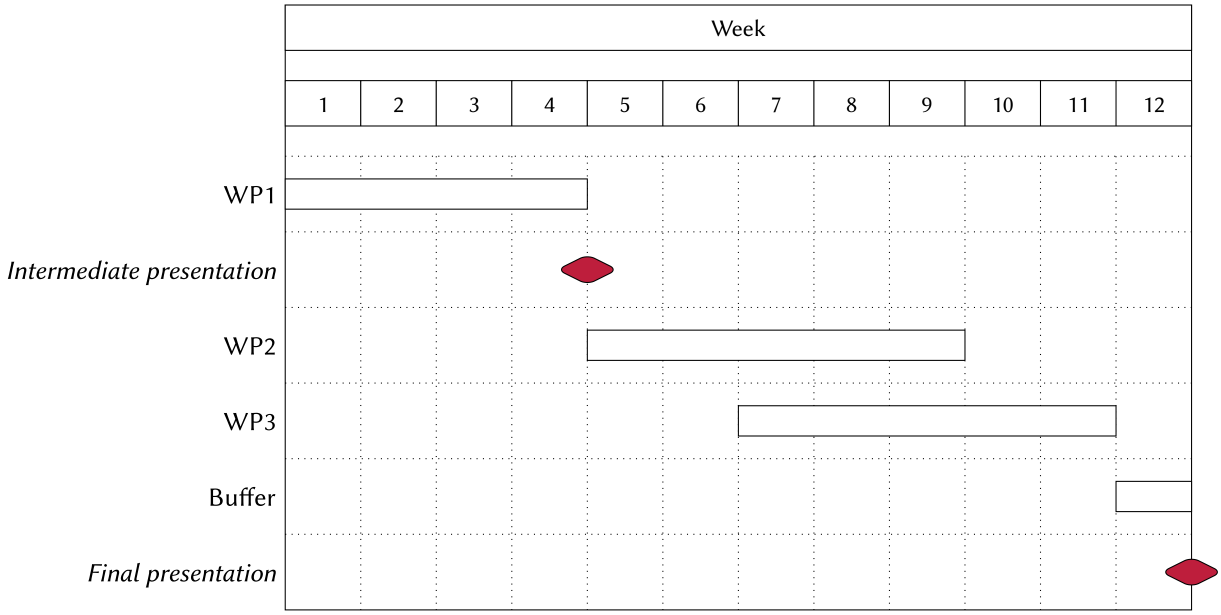
In three to four sentences, the contributions section lists the planned scientific contributions of your work. This naturally includes the findings of your research, but can also include reusable data sets, newly created tools, …
⇒ How do you plan to address your research problem?

3.1 Work Packages

1. **WP1 – First Work Package Headline**
   For each work package, you give a description of one to two sentences.

2. **WP2 – Second Work Package Headline**
   This helps you to structure the necessary steps to achieve your research goal.

3.2 Schedule



References

[TBK09] Thomas Thüm, Don Batory, and Christian Kästner. "Reasoning About Edits to Feature Models". In: *Proc. Int'l Conf. on Software Engineering (ICSE)*. Vancouver, Canada: IEEE, May 2009, pp. 254–264. ISBN: 978-1-4244-3453-4. DOI: 10.1109/ICSE.2009.5070526 (see Page 1).